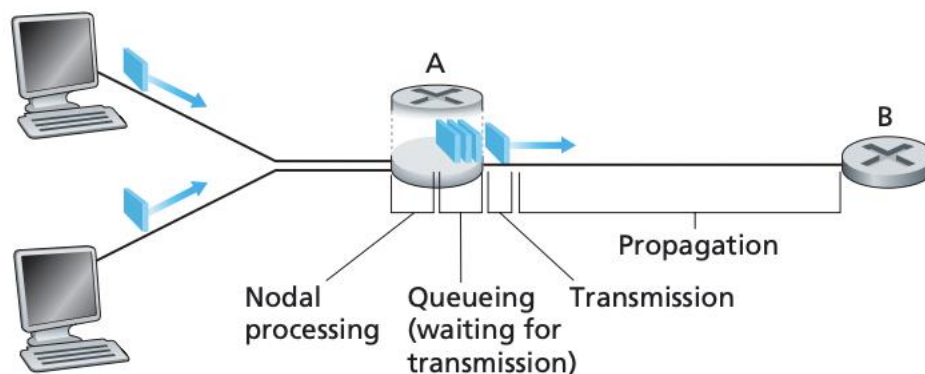


COMP3234 Computer and Communication Networks / ELEC3443 Computer Networks Written Assignment 1

Q1. Consider the network illustrated below. Assume the two hosts on the left of the figure start transmitting packets of 1500 bytes at the same time towards Router B. Suppose the link rates between the hosts and Router A is 4-Mbps. One link has a 6-ms propagation delay and the other has a 2-ms propagation delay. Will queuing delay occur at Router A?



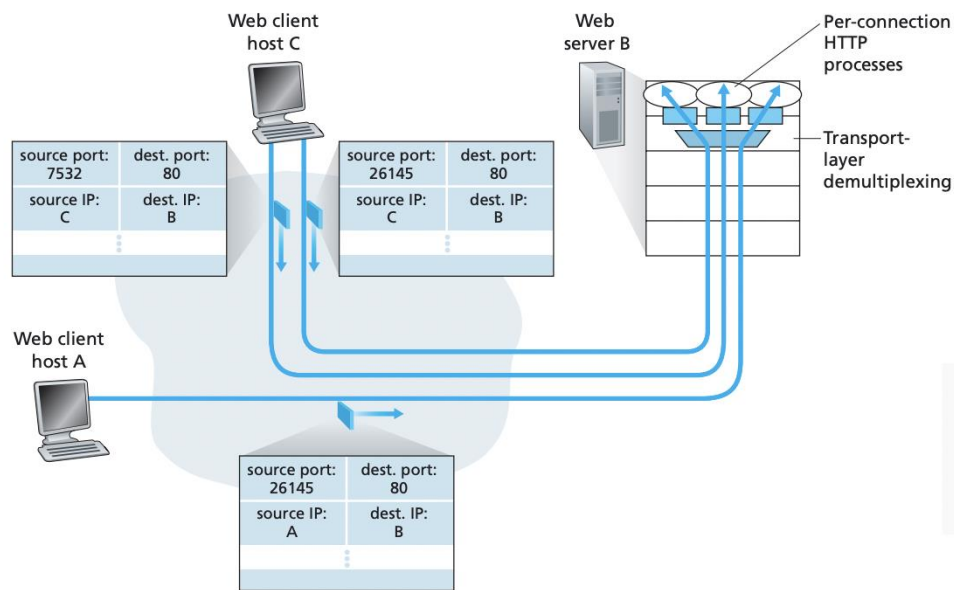
Q2. Consider distributing a file of $F = 17$ Gbits to N peers. The server has an upload rate of $u_s = 40$ Mbps, and each peer has a download rate of $d_i = 4$ Mbps and an upload rate of u . For $N = 20, 200$, and $2,000$ and $u = 400$ Kbps, 600 Kbps, and 4 Mbps, fill in the charts below to show the **minimum distribution time** for each of the combinations of N and u for both client- server distribution and P2P distribution.

Client-server model

	$N=20$	$N=200$	$N=2000$
$u = 400$ Kbps			
$u = 600$ Kbps			
$u = 4$ Mbps			

P2P distribution

	$N=20$	$N=200$	$N=2000$
$u = 400$ Kbps			
$u = 600$ Kbps			
$u = 4$ Mbps			

Q3.

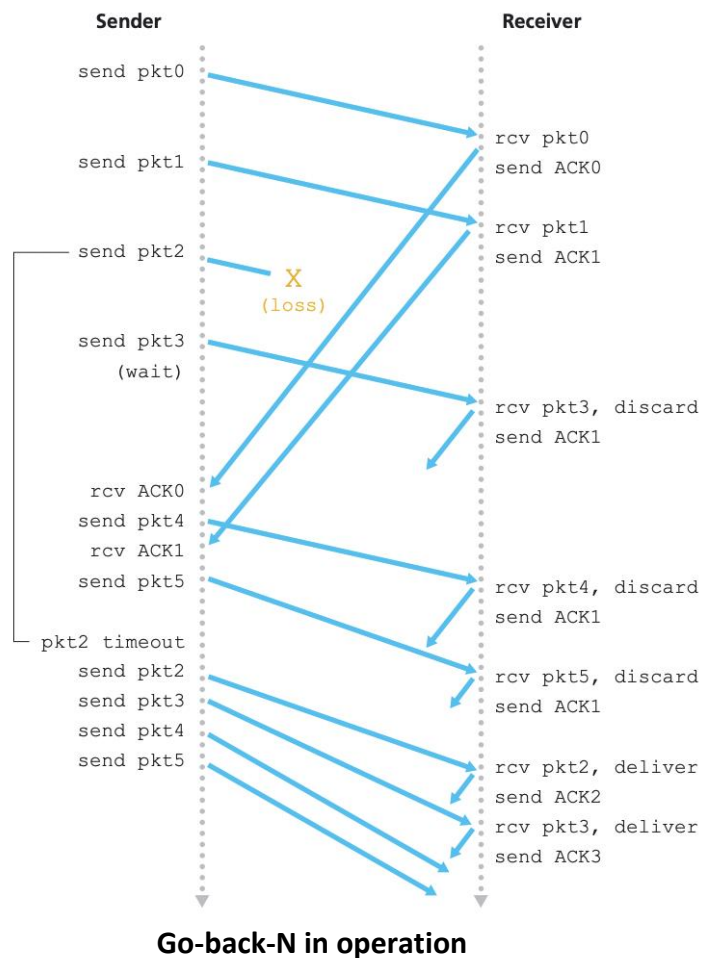
Consider the figure above.

- What are the source and destination port values in the segments flowing from the server back to the clients' processes?
- What are the IP addresses in the network-layer datagrams carrying the transport-layer segments?
(Suppose the IP addresses of the hosts A, B, and C are a, b, c, respectively, where a, b, c are distinct.)

Q4. Consider the GBN protocol with a sender window size of 4 and a sequence number range of 1,024. Suppose that at time t , the next in-order packet that the receiver is expecting has a sequence number of k . Assume that the medium does not reorder messages. Answer the following questions:

- What are the possible sets of sequence numbers inside the sender's window at time t ? Justify your answer.
- What are all possible values of the ACK field in all possible messages currently propagating back to the sender at time t ? Justify your answer.

Q5. Suppose Host A and Host B use a GBN protocol with window size $N = 3$ and a long-enough range of sequence numbers. Assume Host A sends six application messages to Host B and that all messages are correctly received, except for the first acknowledgment and the fifth data segment. Draw a timing diagram (similar to the Figure below), showing the data segments and the acknowledgments sent along with the corresponding sequence and acknowledge numbers, respectively.



Q6. Compare Go-back-N (GBN), Selective Repeat (SR), and TCP (no delayed ACK). Assume that the timeout values for all three protocols are significantly long such that six consecutive data segments and their corresponding ACKs can be received (if not lost in the channel) by the receiving host (Host B) and the sending host (Host A). Suppose Host A sends six data segments to Host B, and the third segment (sent from A) is lost. In the end, all the six data segments have been correctly received by Host B.

- Draw the timing diagram between Host A and Host B for GBN, SR, and TCP, respectively.
- Can you count how many segments has Host A sent in total and how many ACKs has Host B sent in total? What are their sequence numbers?